

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Design of an Interactive Product Calculator using HTML, CSS, and JavaScript

2. AIM

To develop an interactive web page that dynamically calculates the total price of a product based on user inputs for quantity and price.

3. OBJECTIVE

The objectives of this experiment are:

- To design a clean UI using HTML and CSS for a product calculator.
- To use JavaScript to capture input values dynamically.
- To calculate the total price in real-time.
- To format and display the calculated result on the webpage.

4. THEORY

JavaScript enables interactivity in web pages. Event listeners like 'input' can trigger functions immediately whenever the user modifies an input field. By capturing values via `getElementById()`, we can perform calculations and update the Document Object Model (DOM) seamlessly.

5. ALGORITHM / PROCEDURE

1. Create the HTML layout with input fields for product name, quantity, and price.
2. Apply CSS styles to center the calculator and visually differentiate the input and result sections.
3. Define a JavaScript function `calculateTotal()` to retrieve values from inputs.
4. Convert the retrieved string values to Numbers and compute the product (quantity * price).
5. Update the result text content using `.toFixed(2)` for precise currency formatting.
6. Attach 'input' event listeners to the quantity and price fields.

6. CODE EXPLANATION

The script defines a `calculateTotal()` function that reads the string values from the input fields and converts them into numeric format using `Number()`. When the total is calculated, `textContent` is updated to reflect the new

price. Two `addEventListener('input', ...)` calls are bound to the input fields, ensuring the calculation happens synchronously every time a user presses a key or modifies the form.

7. PROGRAM

index_1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Interactive Product Calculator</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f4f7;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
    }

    .calculator {
      background: white;
      width: 350px;
      padding: 30px;
      border-radius: 15px;
      box-shadow: 0 5px 20px rgba(0, 0, 0, 0.15);
    }

    h1 {
      text-align: center;
      color: #333;
      margin-bottom: 25px;
    }

    label {
      display: block;
      margin-top: 15px;
      margin-bottom: 6px;
      font-weight: bold;
      color: #444;
    }

    input {
      width: 100%;
      padding: 10px;
      box-sizing: border-box;
      border: 1px solid #ccc;
      border-radius: 8px;
      font-size: 16px;
    }
  </style>
</head>
<body>
  <h1>Interactive Product Calculator</h1>
  <div class="calculator">
    <div>
      <label>Product Name</label>
      <input type="text" value="Product Name" />
    </div>
    <div>
      <label>Price</label>
      <input type="text" value="Price" />
    </div>
    <div>
      <label>Quantity</label>
      <input type="text" value="Quantity" />
    </div>
    <div>
      <label>Total Price</label>
      <input type="text" value="Total Price" />
    </div>
  </div>
</body>
</html>
```

```

input:focus {
    outline: none;
    border-color: #007bff;
}

.result {
    margin-top: 25px;
    padding: 15px;
    background: #eaf4ff;
    border-radius: 10px;
    text-align: center;
}

.result h2 {
    margin: 0;
    color: #007bff;
}

#totalPrice {
    font-size: 28px;
    font-weight: bold;
    color: #222;
}
</style>
</head>

<body>

    <div class="calculator">

        <h1>Product Calculator</h1>

        <label for="productName">Product Name</label>
        <input type="text" id="productName"
            placeholder="Enter product name">

        <label for="quantity">Quantity</label>
        <input type="number" id="quantity"
            min="0" value="1">

        <label for="price">Price per Product (₹)</label>
        <input type="number" id="price"
            min="0" value="100">

        <div class="result">
            <h2>Total Price</h2>
            <p id="totalPrice">₹100.00</p>
        </div>

    </div>

    <script>

        // Function to calculate total price
        function calculateTotal() {

            // Get values from input fields
            let quantity = Number(

```

```

        document.getElementById("quantity").value
    );

    let price = Number(
        document.getElementById("price").value
    );

    // Calculate total
    let total = quantity * price;

    // Display the result
    document.getElementById("totalPrice").textContent =
        "₹" + total.toFixed(2);
}

// Event listener for quantity
document.getElementById("quantity")
    .addEventListener("input", calculateTotal);

// Event listener for price
document.getElementById("price")
    .addEventListener("input", calculateTotal);

</script>

</body>
</html>

```

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Modify quantity	Total updates instantly	Total price calculates properly based on new quantity	Pass
TC-02	Modify price	Total updates instantly	Total price calculates properly based on new price	Pass

9. OUTPUT

Product Calculator

Product Name

Enter product name

Quantity

1

Price per Product (₹)

100

Total Price

₹100.00

10. RESULT

Thus, an interactive product calculator demonstrating dynamic DOM manipulation, real-time calculation, and event handling was successfully designed and tested.